

Highly sensitive RI and temperature sensor based on cascaded LPGs at near the phase-matching turning point

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ABSTRACT

We demonstrated a highly sensitivity refractive index (RI) and temperature sensor based on cascaded long-period gratings (LPGs) operating at near the phase-matching turning point (PMTP). A series of narrow resonance peaks formed by interference between cladding mode and core mode showed relative high Q-factor comparison with the one of the LPG. The experimental results showed that such fiber sensor have RI sensitivity of $1360\text{nm}/\text{RIU}@1.33$ and $3669.7\text{nm}/\text{RIU}@1.41$ and the temperature sensitivity of $4.079\text{ nm}/^\circ\text{C}$.

Keywords: LPG, Cascaded grating sensor, turning point

1. INTRODUCTION

LPGs could promote light coupling between the core mode and the cladding modes, leaving a series of attenuation peaks in the transmission spectra. The band attenuation can be perturbed by environmentally induced changes in the waveguide dispersion, resulting in high sensitivity to the surrounding refractive index (SRI) and temperature and LPGs has higher sensitivity to environmental perturbation [1]. Several techniques have been proposed to improve the sensitivity of LPG including cladding mode operating at PMTP, LPG inscribed in thin cladding fiber[2], coating with high index film by adjusting the fiber radius[3], and to make LPG operating at the mode transition range [4]. Among of them, the LPG operating at the PMTP is the simplest way to have the higher sensitivity, However, the cladding mode of LPG at the PMTP has a very broad transmission band, which would affect the sensing accuracy. With the change of environmental parameters, the peak at the PMTP gradually splits into two large bandwidth loss peaks, and the distance is very close between two large bandwidth loss peaks, so the small shift of loss peaks is difficult to detect and the

LPG operating at near the PMTP is not suitable for wavelength-modulated sensors[5]. Therefore, when the LPG operates at near the PMTP we need find a method to solve the problem of broad bandwidth and the dual-peak overlap.

A pair of long-period gratings has become the researchers' attention. Because the core and the cladding of the fiber could be realized as the two arms of a Mach-Zehnder interferometer, which would formed a series of narrow bandwidth loss peaks with high Q-factor. Inspired by the narrow bandwidth of the interference loss peak of a LPG pair, the researcher have reported a new high sensitivity sensor in the simulation, which is composed of two identical coated LPGs operating at near the PMTP[6]. Compared with the regular LPG sensors based on dual peaks resonance or the LPG sensors based on mode transition, this LPG pair has a higher sensitivity to SRI[7].

In this paper, we would experimentally demonstrated a cascaded LPGs pair formed MZ-interferometer between forward propagating core and cladding mode and measured the spectrum characteristics of the LPG pair with increasing the refractive index and temperature. Such sensors have showed a high Q-factor sensors, which is suit for high accuracy bio and temperature sensing. we evaluated RI and temperature sensing characteristic of cascaded LPGs pair at the PMTP based fiber sensor under different surrounding refractive index and temperature. Moreover, we calculate the SRI and temperature sensitivity of two narrow bandwidth loss peaks with the highest sensitivity. Our research results show that the high Q-factor LPG pair could be a good solution for measure liquid RI and temperature.

2. FABRICATION AND PRINCIPLE

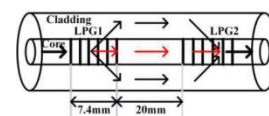


Fig.1 The structure of the LPG pair sensor.

The schematic diagram of the LPG pair is shown in Fig.1, which was fabricated in PS1250/1500 fiber by using point by point inscription method. The period, length of grating and the gap were 154 μ m, 7.4mm and 20mm, respectively. the first LPG was inscribed with 3dB coupling intensity to make sure 50% core mode coupled in to cladding mode and leave 50% core mode inside fiber core. Then, the second the same LPG was cascaded along the fiber with 20mm gap. Such cascaded LPGs formed a mode interference system, in which the interference resonance peaks were appeared in the the transmission spectra of the LPG. The spectrum of LPG1 itself is measured and shown in Fig.2(a), which show that the dual-peak at the center wavelength of the two peak at 1448.2nm and 1752.1nm, respectively, in which the two loss peaks have wide bandwidth result to they are hardly to evaluate clearly. After inscribed the second LPG, the spectrum of the cascaded LPGs pair is shown in Fig.2.(b), which showed that the transmission spectrum was composed of a series of narrow bandwidth loss peaks and we mark them on the peak 1 to 7. The solid curve and the dotted curve show the spectrum of the LPG pair at refractive index 1 and 1.33, in which the peak 4 and peak5 not appear until the increase of the refractive index and then they will shift in the opposite direction.

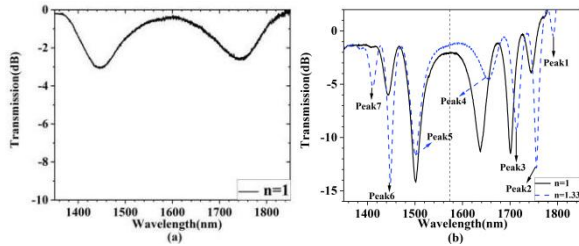


Fig.2(a) The spectrum of the LPG1.(b) The spectrum of the LPG pair operating at the PMTP.

However, it is evident that high Q-factor is necessary for smaller resonance shifts. To improve the Q-factor can achieve accurate measure the change of transmission spectrum on different SRIs and temperature by The Q-factor of the cascaded LPGs pair coupled to a single waveguide is given by[8]

$$Q = \frac{2\pi n_{\text{eff}} L}{\lambda_0 \cdot 2 \arccos \left\{ \frac{[1 - 4t/\tau + (t/\tau)^2]/(-2t/\tau)}{1} \right\}} = \frac{\lambda_0}{\Delta\lambda_{3\text{dB}}} \quad (1)$$

where n_{eff} is the effective index of refraction of the propagating mode in the LPG pair, λ_0 is the resonance wavelength, L is the length of the LPG pair is the field transmission coefficient at the waveguide resonator coupling, and $\tau = \exp(-\alpha L/2)$ is the attenuation of the field per round trip for a LPG pair resonator with attenuation constant α , a reasonably noticeable shift of the resonance wavelength should be of the order of $\Delta\lambda_{\text{detect}} \approx 1/2\Delta\lambda_{3\text{dB}}$ [8]. The Q-factor of the regulation dual-peak is to be estimated $\sim 1511\text{nm}/\sim 76.9\text{nm} \cong 19.6$

and $\sim 1745.8\text{nm}/\sim 89.7\text{nm} \cong 19.4$, and the Q-factor of the peak 4 and 5 by cascaded LPGs is measured to be $\sim 1512\text{nm}/\sim 38\text{nm} \cong 39.8$ and $\sim 1639\text{nm}/\sim 46\text{nm} \cong 35.6$, which show the cascaded LPG has almost two times higher Q-factor than the traditional LPGs operating at PMTP.

3. EXPERIMENTAL AND RESULTS

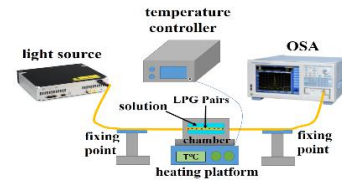


Fig.3 Schematic for the characterization of LPG towards surrounding refractive index and temperature.

To investigate the sensing characteristic of proposed fiber sensor, we have subjected the sensor into different refractive index from 1.33 to 1.41 and also applied the TEC to measure the temperature sensing performance. As shown in Fig.3, the effective zone of sensor is immersed into different refractive index liquid solution which were made by glycerin solution with different water proportioning and the refractive index value was measured by an Abbe refractometer.

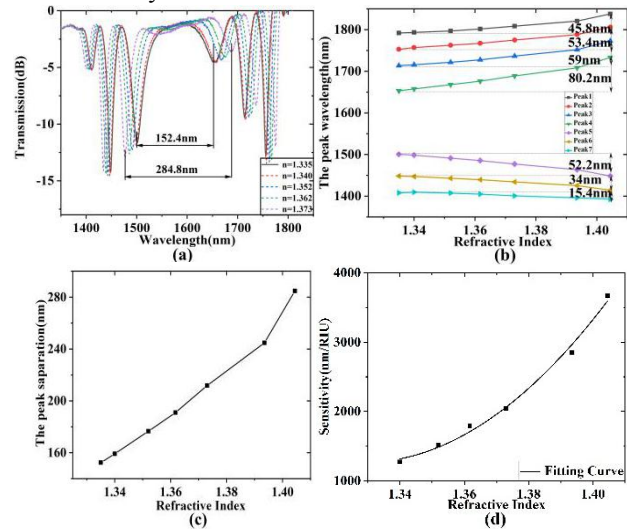


Fig.4 (a) Spectrum of the LPG pair under different SRIs.(b) Change of the peak wavelength of different loss peaks on the SRI.(c) the peak separation between peak 4 and 5 with different SRI.(d) RI sensitivity of different SRIs.

The dependence of the spectrum on the SRI is shown in Fig.4(a), with increasing of the SRI, the interference peaks of dual-peak shift in opposite direction, which the wavelength of peaks at the left of waveguide dispersion turning point shift to the short wavelength and the right one move to the longer wavelength. The seven loss peaks were observed as shown in Fig.4(b), we can see the peak4 and peak5 have the fastest shift of all, the peak4 changes 80.2 nm in the long wavelength direction and the peak5 changes 52.2 nm in the short

wavelength direction. the wavelength spacing of the peak4 and peak5 changes from 152.4nm to 284.8nm with total 142.3nm wavelength spacing change as the index value from 1.335-1.401, see in Fig.4(c). according to the RI sensing results in Fig. 4(c), we have calculated the sensitivity was calculated by the peak shift of the peak4 and 5 with different refractive indices and plotted and fitted in Fig.4(d). As it shown in Fig.4(d), the RI sensitivity is not constant, which is gradually increasing with the increasing of the SRI; the measurement results show that the RI sensitivity is around 1360nm/RIU at the index of 1.33.

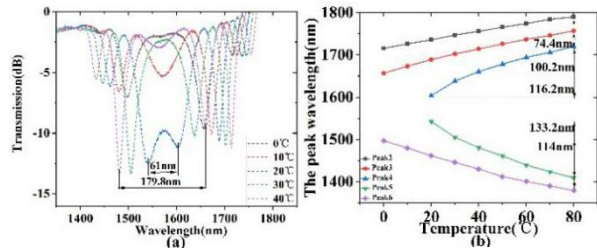


Fig.5(a). Spectrum of the LPG pair under different temperature. (b) Change of different peaks wavelength on the temperature.

We also investigated the temperature sensing performance by using a TEC instrument see in Fig. 3. The spectrum of the LPG pair under different temperature is shown in Fig.5(a) with the temperature changing from 0°C to 40°C. As we mentioned before, there were only 5 the resonance peaks when the sensor was in air. The temperature sensing performance of each peak was shown in Fig.5(b), and the temperature sensitivity of peak 2, peak 3, peak4, peak 5 and peak 6 were 0.885nm/°C, 1.085nm/°C, 1.605nm/°C, 1.917nm/°C, and 1.221/°C, respectively, in which the peak 4 and peak 5 have the largest wavelength shift and different shifting direction as temperature increasing from 20°C to 80°C, and were 116.2nm and -133.2nm, respectively.

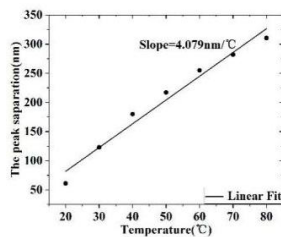


Fig.6 Linear fitting curves of the peak separation between peak 4 and 5 with different temperature.

As it shown in Fig.5(b), the peak 4 and peak 5 have the relative higher absolute temperature sensitivity and different wavelength shifting direction. The temperature measuring sensitivity could be doubled by monitoring the peak separation between peak 4 and peak 5, and the result was plotted in Fig.6, which has shown a linear temperature response with the sensitivity of 4.079nm/°C. With the 0.02 minimum wavelength resolution, the temperature sensing accuracy is around $5.5 \times 10^{-3}^\circ\text{C}$.

4. CONCLUSION

In conclusion, we have demonstrated and fabricated a high sensitivity RI and temperature sensor based on the cascaded LPGs, which has showed a higher Q-factor sensing peak to achieve the higher measuring accuracy compare with the single LPG based sensor. The LPG was designed with 154μm period to ensure the grating operating at the near PMPT. The experiment results have shown the proposed fiber sensor has a up to 3669.7nm/RIU at 1.41 and the temperature is around 4.079nm/°C. Due to the different sensitivity of the free spectra range, the peak shift and spacing of cascaded LPGs, it would be possible to achieve simultaneously measuring for RI and temperature. Based on the advantage of high sensitivity, such sensor will be potentially applied in biochemical sensing field.

5. ACKNOWLEDGMENTS

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